Solution

(a) Electric potential at point P is given by

$$V_P = \frac{1}{4\pi\varepsilon_\circ}\frac{q}{r_P} = \frac{9\times10^9\times9\times10^{-6}}{10} = 8.1\times10^3 V$$

Electric potential at point Q is given by

$$V_Q = \frac{1}{4\pi\varepsilon_\circ}\frac{q}{r_Q} = \frac{9\times10^9\times9\times10^{-6}}{16} = 5.06\times10^3 V$$

Note that the electric potential at point Q is less than the electric potential at point P. If we put a positive charge at P, it moves from P to Q. However if we place a negative charge at P it will move towards the charge +9μC.

The potential difference between the points P and Q is given by

$$\Delta V = V_P - V_Q = +3.04\times10^3 V$$

(b) Suppose we replace the charge +9 μC by -9 μC, then the corresponding potentials at the points P and Q are,

$$V_P = -8.1\times10^3 V,\ V_Q = -5.06\times10^3 V$$

Note that in this case electric potential at the point Q is higher than at point P.

The potential difference or voltage between the points P and Q is given by

$$\Delta V = V_P - V_Q = -3.04\times10^3 V$$

(c) The electric potential V at a point P due to some charge is defined as the work done by an external force to bring a unit positive charge from infinity to P. So to bring the q amount of charge from infinity to the point P, work done is given as follows.

$$W = qV$$

$$W_Q = 2\times10^{-6}\times5.06\times10^3 J = 10.12\times10^{-3} J.$$

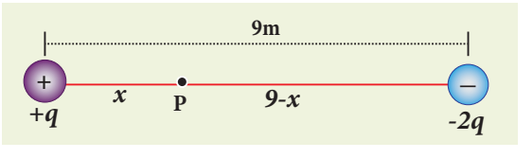
EXAMPLE 1.13

Consider a point charge +q placed at the origin and another point charge -2q placed at a distance of 9 m from the charge +q. Determine the point between the two charges at which electric potential is zero.

Solution

According to the superposition principle, the total electric potential at a point is equal to the sum of the potentials due to each charge at that point.

Consider the point at which the total potential zero is located at a distance x from the charge +q as shown in the figure.



The total electric potential at P is zero.

$$V_{tot} = \frac{1}{4\pi\varepsilon_\circ}\left(\frac{q}{x} - \frac{2q}{(9-x)}\right) = 0$$

which gives $\frac{q}{x} = \frac{2q}{(9-x)}$

or $\frac{1}{x} = \frac{2}{(9-x)}$

Hence, $x = 3$ m

1.5.3 Electrostatic potential at a point due to an electric dipole

Consider two equal and opposite charges separated by a small distance 2a as shown in Figure 1.26. The point P is located at a distance r from the midpoint of the dipole. Let θ be the angle between the line OP and dipole axis AB.

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